

ELEC2134 : Circuits and Signals

PART A (Week 1 to Week 5)

Transform Methods

- Periodic signals and Fourier series
- Aperiodic signals and Fourier transforms
- Laplace transforms of signals and circuits

System Analysis

- Linear systems and convolution
- Impulse response
- Frequency response and transfer functions
- Steady-state analysis of linear systems
- Transient analysis of 1st and 2nd systems
- Circuit analysis using transform methods

PART B (Week 7 to Week 10)

AC Circuit Analysis

- AC circuits with sinusoidal inputs
- Use of phasors and complex impedance in AC circuit analysis
- AC power: real, reactive and apparent power
- Power factor (leading/lagging)
- Series and parallel resonance
- Transformers and coupled coils
- Application of network theorems: Two port networks

ELEC2134 : Circuits and Signals (PART A)

Topic 3: Transform Methods:

Laplace Transforms of Signals and Circuits

Laplace transforms convert circuit equations into the s-domain, where differential equations become simple **algebraic equations**, making it much easier to analyse transient and steady-state responses of circuits.

Example: Laplace Transform of Circuits

This means analysing **RLC circuits** using Laplace transforms instead of differential equations:

Time domain

$$L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int_0^t i(\tau) d\tau = v(t)$$

Laplace or s-domain

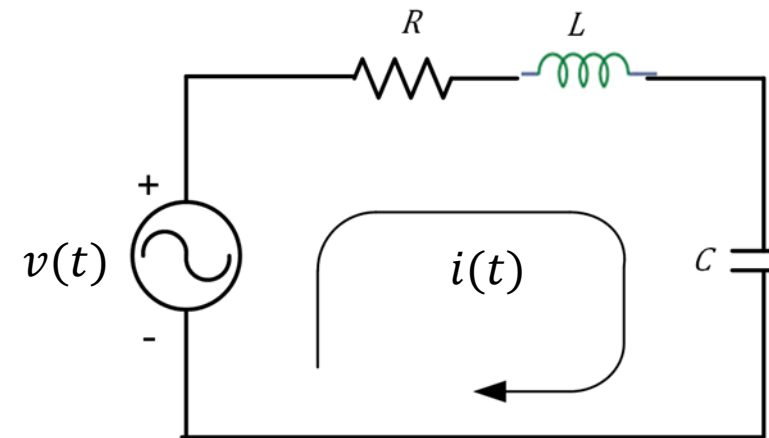
$$\left(sL + R + \frac{1}{sC} \right) I(s) = V(s)$$

$$I(s) = \frac{V(s)}{sL + R + \frac{1}{sC}}$$

Important: The algebraic form above assumes zero initial conditions.

Nonzero initial capacitor voltage or inductor current introduces additional source terms in the s-domain model.

S - domain : Complex frequency domain {s is a complex variable; $s = \sigma + j\omega$ }



RLC circuit

For example, Laplace transforms let you:

- Convert differential equations to algebraic equations
- Handle initial conditions easily
- Analyse transient and steady-state responses
- Obtain transfer functions
- Obtain the Fourier transform as a special case of the Laplace transform (when $s = j\omega$)

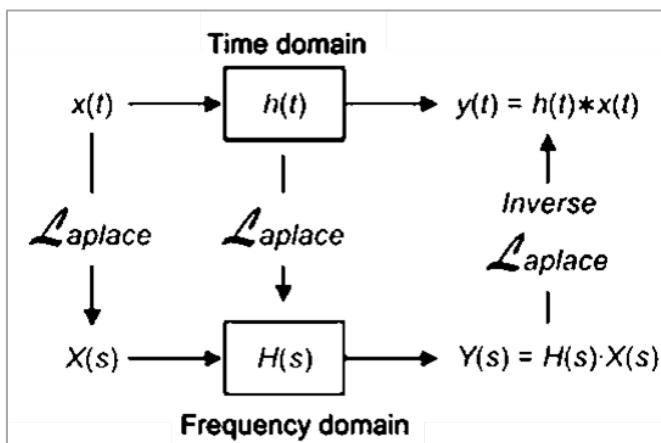
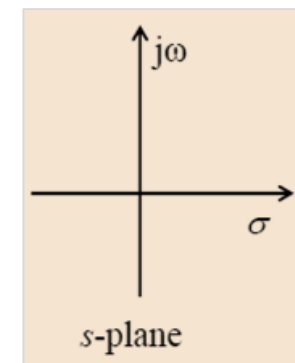
Introduction to Laplace Transform

- The Laplace transform is an integral transformation that maps a function $f(t)$ from the time domain to the complex frequency domain, yielding $F(s)$.

- Definition: $\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st}dt$ [one sided Laplace transform assumes $f(t) = 0, t < 0$]

where s is a complex variable and given by $s = \sigma + j\omega$; σ - exponential damping factor ω - frequency

- It is interesting to know that,
 $s = j\omega$ for sinusoidal AC signals (Fourier/frequency axis)
 $s = \sigma$ for exponential signals: purely real exponential weighting
 $s = \sigma + j\omega$ for exponential sinusoidal signals: exponential weighting plus oscillation



Fourier told us *what frequencies are present*.

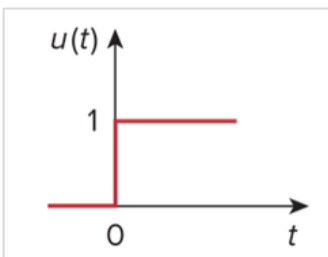
Laplace also lets us handle *growth/decay, transients and initial energy*.

Introduction to Laplace Transform

Example 1

Determine the Laplace transform of each of the following functions:

(a) $u(t)$ (Unit-step function)



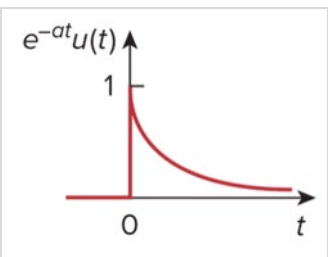
$$\begin{aligned} U(s) = \mathcal{L}[u(t)] &= \int_0^{\infty} 1 e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} \\ &= -\frac{1}{s}(0) + \frac{1}{s}(1) = \frac{1}{s} \end{aligned}$$

The **inverse Laplace transform** converts a function from the s-domain back to the time domain.

$$\mathcal{L}^{-1}[F(s)] = f(t)$$

$$\mathcal{L}^{-1}\left[\frac{1}{s}\right] = u(t)$$

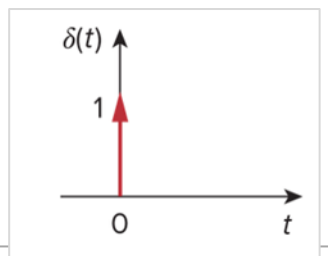
(b) $f(t) = e^{-at} u(t)$, $a \geq 0$ (Exponential function)



$$\begin{aligned} F(s) = \mathcal{L}[e^{-at} u(t)] &= \int_0^{\infty} e^{-at} e^{-st} dt \\ &= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} = \frac{1}{s+a} \end{aligned}$$

$$\mathcal{L}^{-1}\left[\frac{1}{s+a}\right] = e^{-at} u(t)$$

(c) $\delta(t)$ (An impulse)



$$\mathcal{L}[\delta(t)] = \int_0^{\infty} \delta(t) e^{-st} dt = e^{-0} = 1$$

Laplace Transform(LT) vs Fourier Transform (FT)

Relationship between LT and FT

- Laplace transform (being $s = \sigma + j\omega$):

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

- Fourier Transform

$$FT \{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

- **Relationship between LT and FT:**

For a function $f(t)$ that is zero for $t < 0$ and satisfies

$$\int_0^{\infty} |f(t)| dt < \infty$$

the two transforms are related by $F(\omega) = F(s)|_{s=j\omega}$.

- **Conclusion:**

The Fourier transform is a special case of the Laplace transform with

$$s = j\omega \quad (\sigma = 0)$$

Region of Convergence (ROC)

- When we take the Laplace transform,

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt, \quad s = \sigma + j\omega,$$

the integral does not converge for all values of s .

- The Region of Convergence (ROC) is the set of values of s in the complex s -plane for which the integral converges (is finite).

- Because of the exponential term:

$$e^{-st} = e^{-\sigma t} e^{-j\omega t}$$

❖ $e^{-j\omega t} \rightarrow$ oscillates (does not decay)

❖ $e^{-\sigma t} \rightarrow$ controls growth or decay

So:

❖ If $\sigma > 0$: exponential decays $\rightarrow$ helps convergence

❖ If $\sigma < 0$: exponential grows $\rightarrow$ may cause divergence

- The ROC is the range of σ values for which the integral converges.
- The Fourier transform (FT) exists only if the ROC includes the imaginary axis.

Laplace Transform Properties - Linearity

1. Linearity

- If $F_1(s)$ and $F_2(s)$ are, respectively the Laplace Transforms of $f_1(t)$ and $f_2(t)$, then

$$\mathcal{L}\{a_1 f_1(t) + a_2 f_2(t)\} = a_1 F_1(s) + a_2 F_2(s)$$

Example 2

- Find Laplace transform of $f(t) = \cos(\omega t)u(t)$

$$\mathcal{L}\{\cos(\omega t) u(t)\} = \mathcal{L}\left\{\frac{1}{2}e^{j\omega t} + \frac{1}{2}e^{-j\omega t}\right\} = \frac{1}{2}\mathcal{L}[e^{j\omega t}] + \frac{1}{2}\mathcal{L}[e^{-j\omega t}]$$

$$= \frac{1}{2}\mathcal{L}[e^{j\omega t}] + \frac{1}{2}\mathcal{L}[e^{-j\omega t}] = \frac{1}{2}\left(\frac{1}{s-j\omega} + \frac{1}{s+j\omega}\right)$$

$$\mathcal{L}\{\cos(\omega t) u(t)\} = \frac{s}{s^2 + \omega^2}$$

- Similarly $\mathcal{L}\{\sin(\omega t) u(t)\} = \frac{\omega}{s^2 + \omega^2}$

← We have $\mathcal{L}\{e^{-at}\} = \left[\frac{1}{s+a}\right]$

Laplace Transform Properties – Scaling , Time shift and Frequency Shift

2. Scaling

- If $F(s)$ is the Laplace Transform of $f(t)$ then

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

Example 3

- Find Laplace transform of $f(t) = \cos(2\omega t)u(t)$

$$\text{We have } \mathcal{L}\{\cos(\omega t)u(t)\} = \frac{s}{s^2 + \omega^2}$$

$$\text{Therefore, } \mathcal{L}\{\cos(2\omega t)u(t)\} = \frac{1}{2} \left[\frac{\left(\frac{s}{2}\right)}{\left(\frac{s}{2}\right)^2 + \omega^2} \right] = \frac{s}{s^2 + 4\omega^2}$$

4. Frequency shift

- If $F(s)$ is the Laplace Transform of $f(t)$ then

$$\mathcal{L}\{e^{-at}f(t)u(t)\} = F(s + a)$$



3. Time shift

- If $F(s)$ is the Laplace Transform of $f(t)$ then

$$\mathcal{L}\{f(t - a)u(t - a)\} = e^{-as}F(s)$$

Example 4

- Find Laplace transform of
 $f(t) = \cos(\omega(t - a))u(t - a)$

$$\text{We have } \mathcal{L}\{\cos(\omega t)u(t)\} = \frac{s}{s^2 + \omega^2}$$

$$\text{Therefore, } \mathcal{L}\{\cos(\omega(t - a))u(t - a)\} = e^{-as} \frac{s}{s^2 + \omega^2}$$

Example 5

- Find Laplace transform of
 $f(t) = e^{-at}\cos(\omega t)u(t)$

$$\text{We have } \mathcal{L}\{\cos(\omega t)u(t)\} = \frac{s}{s^2 + \omega^2}$$

$$\text{Therefore, } \mathcal{L}\{e^{-at}\cos(\omega t)u(t)\} = \frac{s + a}{(s + a)^2 + \omega^2}$$

Laplace Transform Properties – Time Differentiation

5. Time differentiation

- If $F(s)$ is the Laplace Transform of $f(t)$ then the Laplace Transform of its derivative is

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0^-)$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$$

$f(0^-)$: It is the value of the signal just *before* $t = 0$. This is the initial value of the signal at $t = 0$ *before* any input or switching happens.

0^- means immediately before switching/input application.

Note: $f(0^-)$ represents the stored energy before switching

- Capacitor voltage $v_C(0^-)$
- Inductor current $i_L(0^-)$

$$\mathcal{L}\left[\frac{d^n f}{dt^n}\right] = s^n F(s) - s^{n-1}f(0^-) - s^{n-2}f'(0^-) - \dots - s^0 f^{n-1}(0^-)$$

Example 6

- If $\mathcal{L}\{f(t) = \cos(\omega t) u(t)\} = \frac{s}{s^2 + \omega^2}$; use Laplace transform properties to determine $\mathcal{L}\{\sin(\omega t) u(t)\}$

Note: Use the derivative property or known Laplace transform pairs.

$$\begin{aligned}\mathcal{L}\{\sin(\omega t) u(t)\} &= \frac{-1}{\omega} \mathcal{L}\{f'(t)\} = \frac{-1}{\omega} [sF(s) - f(0^-)] \\ &= \frac{-1}{\omega} \left[s \frac{s}{s^2 + \omega^2} - 1 \right] = \frac{\omega}{s^2 + \omega^2}\end{aligned}$$

Laplace Transform Properties – Time Integration and Frequency Differentiation

6. Time Integration

- If $F(s)$ is the Laplace Transform of $f(t)$ then the Laplace Transform of its integral is

$$\mathcal{L}\left[\int_0^t f(t)dt\right] = \frac{1}{s}F(s)$$

Example 7

- If $\mathcal{L}\{u(t)\} = \frac{1}{s}$; use Laplace transform properties to determine $\mathcal{L}\{t\}$.

We have, $\int_0^t u(t)dt = t$; Taking LT on both sides, we

obtain $\mathcal{L}\left[\int_0^t u(t)dt\right] = \mathcal{L}[t] = \frac{1}{s}U(s)$ {using the above property}

$$= \frac{1}{s}\left(\frac{1}{s}\right) = \frac{1}{s^2}$$

Note: In general $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$

7. Differentiation in the s-domain (multiplication by t)

- If $F(s)$ is the Laplace Transform of $f(t)$ then the derivative with respect to s is

$$\mathcal{L}[t f(t)] = -\frac{dF(s)}{ds}$$

Also

$$t^n f(t) = (-1)^n \frac{d^n F(s)}{ds^n}$$

Example 8

- Find Laplace transform of $t e^{-at}$.
- Let $f(t) = e^{-at}$; $\therefore F(s) = \frac{1}{s+a}$

$$\mathcal{L}[t e^{at}] = -\frac{dF(s)}{ds} = -\frac{d}{ds}\left(\frac{1}{s+a}\right) = \frac{1}{(s+a)^2}$$

Laplace Transform Properties – Frequency Integration (Division by t)

8. Integration in the s-domain (Division by t)

- If $F(s)$ is the Laplace Transform of $f(t)$ then the integration with respect to s is

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)ds$$

Example 9

- Find Laplace transform of $\frac{1-e^{-at}}{t}$, $a > 0$.

$$\text{Let } f(t) = e^{-at}, a > 0; \therefore F(s) = \frac{1}{s} - \frac{1}{s+a}$$

$$\begin{aligned}\mathcal{L}\left[\frac{e^{-at}}{t}\right] &= \int_s^\infty F(s)ds \\ &= \int_s^\infty \frac{1}{s} - \frac{1}{s+a} ds = 0 - (\log(s) - \log(s+a)) = \log \frac{s+a}{s}\end{aligned}$$

Initial and Final Value Theorem

- Initial and final values of a function can be found directly from its Laplace transform

$$\text{Initial Value Theorem: } f(0) = \lim_{s \rightarrow \infty} sF(s)$$

$$\text{Final Value Theorem: } f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

For example, if $f(t) = e^{-2t} \sin(5t) u(t)$, then

$$F(s) = \mathcal{L}[f(t)] = \frac{5}{(s+2)^2 + 5^2};$$

$$\therefore f(\infty) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{5s}{s^2 + 4s + 29} = 0$$

The Final Value Theorem is applicable only when the system has a finite final value

For rational $F(s)$, the poles of $sF(s)$ must lie in the open left-half plane (apart from simple origin case).

Inverse Laplace Transform

Example 10

- Find the inverse Laplace transform of

$$F(s) = \frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}$$

Solution

$$f(t) = \mathcal{L}^{-1}\left(\frac{3}{s}\right) - \mathcal{L}^{-1}\left(\frac{5}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{6}{s^2+4}\right)$$

$$\therefore f(t) = (3 - 5e^{-t} + 3 \sin(2t)) \quad t \geq 0$$

$$f(t) = (3 - 5e^{-t} + 3 \sin(2t))u(t)$$

Basic Laplace Transform Pairs

Time Domain $f(t)$, $t \geq 0$ Laplace Transform $F(s)$

$$u(t) \quad \frac{1}{s}$$

$$e^{-at} \quad \frac{1}{s+a}$$

$$\sin \omega t \quad \frac{\omega}{s^2 + \omega^2}$$

$$\cos \omega t \quad \frac{s}{s^2 + \omega^2}$$

$$e^{-at} f(t) \quad F(s+a)$$

$$t^n \quad \frac{n!}{s^{n+1}}$$

$$te^{-at} \quad \frac{1}{(s+a)^2}$$

$$e^{-at} \sin \omega t \quad \frac{\omega}{(s+a)^2 + \omega^2}$$

$$e^{-at} \cos \omega t \quad \frac{s+a}{(s+a)^2 + \omega^2}$$

$$e^{-bt} t^n \quad \frac{n!}{(s+b)^{n+1}}$$

$$\mathcal{L}\left\{\frac{d^k f(t)}{dt^k}\right\} = s^k F(s) - s^{k-1} f(0^-) - s^{k-2} f'(0^-) - \dots - f^{(k-1)}(0^-)$$

Inverse Laplace Transform

Example 11

- Find the inverse Laplace transform of

$$F(s) = \frac{s^2 + 12}{s(s+2)(s+3)}$$

Solution

First find the partial fractions:

$$\frac{s^2 + 12}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

$$F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

$$f(t) = \mathcal{L}^{-1}\left(\frac{2}{s}\right) - \mathcal{L}^{-1}\left(\frac{8}{s+2}\right) + \mathcal{L}^{-1}\left(\frac{7}{s+3}\right)$$

$$\therefore f(t) = 2u(t) - 8e^{-2t} + 7e^{-3t} \quad t \geq 0$$

Residue Method for Finding the Values of A, B, and C:

$$A = sF(s) \Big|_{s=0} = \frac{s^2 + 12}{(s+2)(s+3)} \Big|_{s=0} = \frac{12}{(2)(3)} = 2$$

$$B = (s+2)F(s) \Big|_{s=-2} = \frac{s^2 + 12}{s(s+3)} \Big|_{s=-2} = \frac{4+12}{(-2)(1)} = -8$$

$$C = (s+3)F(s) \Big|_{s=-3} = \frac{s^2 + 12}{s(s+2)} \Big|_{s=-3} = \frac{9+12}{(-3)(-1)} = 7$$

Algebraic method:

$$s^2 + 12 = A(s^2 + 5s + 6) + B(s^2 + 3s) + C(s^2 + 2s)$$

Equating the coefficients of like powers of s gives

$$\text{Constant: } 12 = 6A \implies A = 2$$

$$s: \quad 0 = 5A + 3B + 2C \implies 3B + 2C = -10$$

$$s^2: \quad 1 = A + B + C \implies B + C = -1$$

$$\text{Thus } A = 2, B = -8, C = 7$$

Solving differential equations using Laplace Transform

Example 12

- Use the Laplace transform to solve the following differential equation:

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

Given : $v(0) = 1$; $v'(0) = -2$;

Laplace transform of its derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$$

Solution

Taking the Laplace transform of each term in the given differential equation yields:

$$[s^2V(s) - sv(0) - v'(0)] + 6[sV(s) - v(0)] + 8V(s) = \frac{2}{s}$$

Substituting $v(0) = 1$; $v'(0) = -2$, we obtain

$$(s^2 + 6s + 8)V(s) = s + 4 + \frac{2}{s} = \frac{s^2 + 4s + 2}{s} \Rightarrow V(s) = \frac{\frac{1}{4}}{s} + \frac{\frac{1}{2}}{s+2} + \frac{\frac{1}{4}}{s+4}$$

Taking the inverse Laplace transform, we obtain: $v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t)$

Three Steps:

1. Change the circuit from the time domain to the s-domain.
2. Solve the circuit in the s-domain.
3. Take the inverse Laplace transform to return to the time domain.

Solving differential equations using Laplace Transform

Example 13

- Use the Laplace transform to solve the following differential equation:

$$\frac{dy}{dt} + 5y(t) + 6 \int_0^t y(\tau) d\tau = u(t),$$

Given : $y(0) = 2$;

Solution

Taking the Laplace transform of each term in the given differential equation yields:

$$[sY(s) - y(0)] + 5Y(s) + \frac{6}{s}Y(s) = \frac{1}{s}$$

Substituting $y(0) = 2$; we obtain

$$Y(s)(s^2 + 5s + 6) = 1 + 2s \quad Y(s) = \frac{2s + 1}{(s + 2)(s + 3)} \quad Y(s) = \frac{2s + 1}{(s + 2)(s + 3)} = \frac{A}{s + 2} + \frac{B}{s + 3}$$
$$Y(s) = \frac{-3}{s + 2} + \frac{5}{s + 3}$$

Taking the inverse Laplace transform, we obtain: $y(t) = (-3e^{-2t} + 5e^{-3t}) u(t)$

Laplace transform of its derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$$

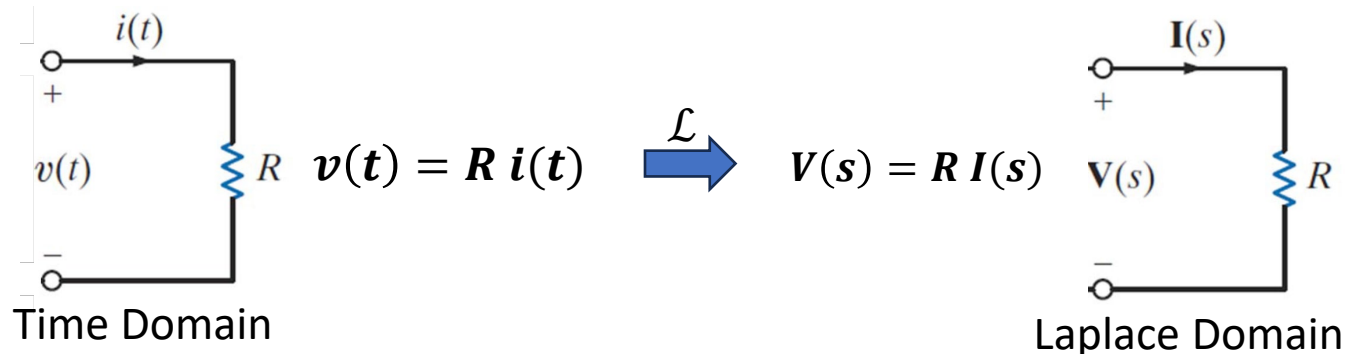
$$\mathcal{L}\left[\int_0^t f(t)dt\right] = \frac{1}{s}F(s)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

1. Write the differential equation
2. Take LT of both sides
3. Insert initial conditions
4. Solve algebraically for $Y(s)$
5. Partial fractions
6. Inverse LT
7. Check initial/final behaviour

Laplace-Domain Circuit Models

Resistive circuit



Laplace transform of its derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-)$$

$$\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0^-) - f'(0^-)$$

$$\mathcal{L}\left[\int_0^t f(t)dt\right] = \frac{1}{s}F(s)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

Inductive circuit

Time Domain

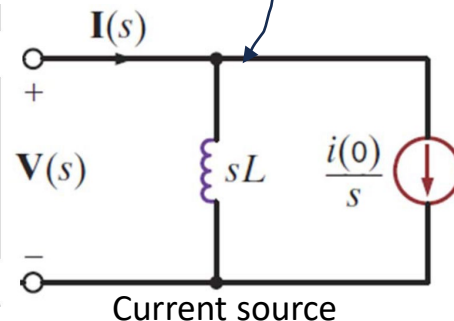
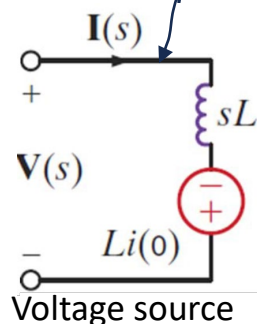
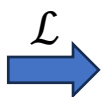
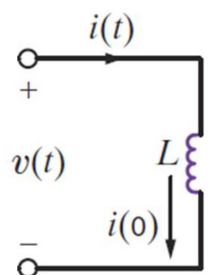
$$v(t) = L \frac{di(t)}{dt}$$

$$i(t) = \frac{1}{L} \int_0^t v(x) dx + i(0)$$

s - Domain

$$\mathbf{V}(s) = sL\mathbf{I}(s) - Li(0)$$

$$\mathbf{I}(s) = \frac{\mathbf{V}(s)}{sL} + \frac{i(0)}{s}$$



Laplace-Domain Circuit Models

Capacitive circuit

Time Domain

$$v(t) = \frac{1}{C} \int_0^t i(x) dx + v(0)$$

$$i(t) = C \frac{dv(t)}{dt}$$

s - Domain

$$\mathbf{V}(s) = \frac{\mathbf{I}(s)}{sC} + \frac{v(0)}{s}$$

$$\mathbf{I}(s) = sC\mathbf{V}(s) - Cv(0)$$

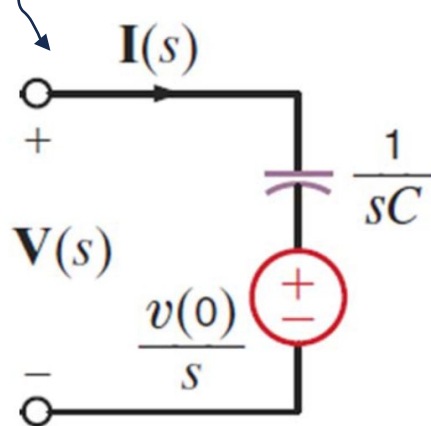
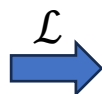
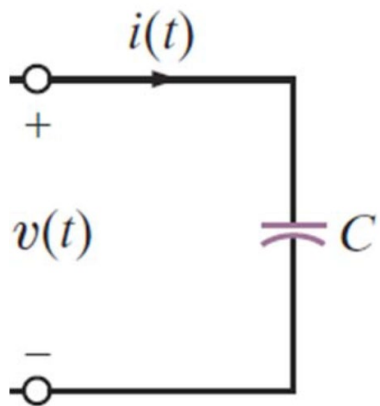
Laplace transform of its derivative

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-)$$

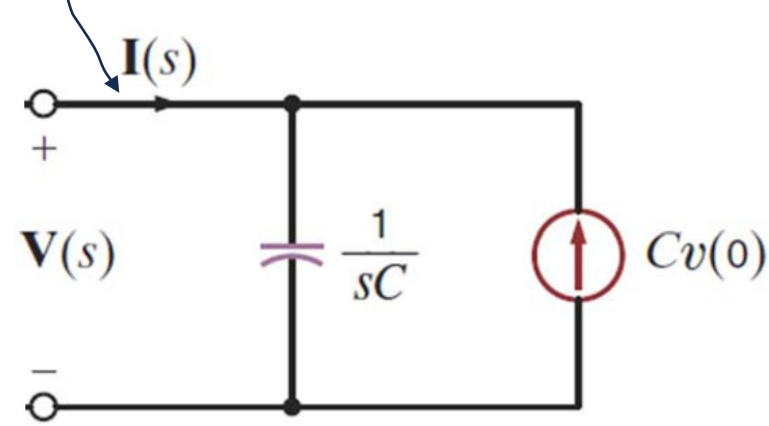
$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$$

$$\mathcal{L}\left[\int_0^t f(t)dt\right] = \frac{1}{s}F(s)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$



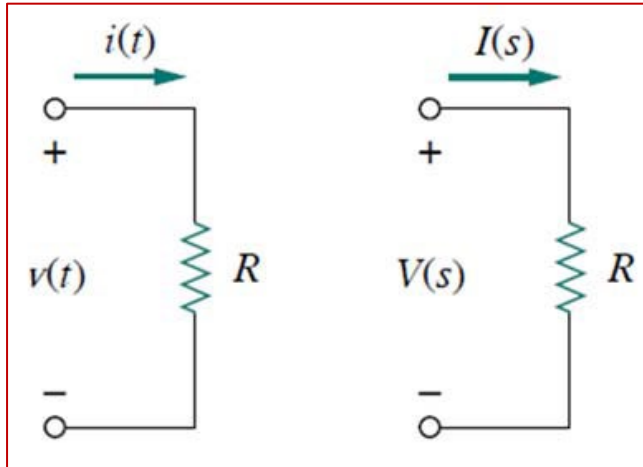
Voltage source



Current source

Laplace-Domain Circuit Models – Assuming zero initial conditions

Resistor



$$V(s) = R I(s)$$

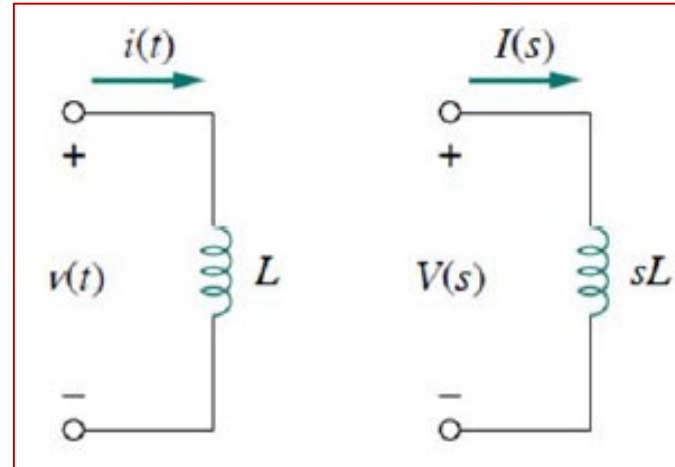
Impedance:

$$Z(s) = \frac{V(s)}{I(s)} = R$$

Admittance:

$$Y(s) = \frac{I(s)}{V(s)} = \frac{1}{R}$$

Inductor



$$V(s) = sL I(s)$$

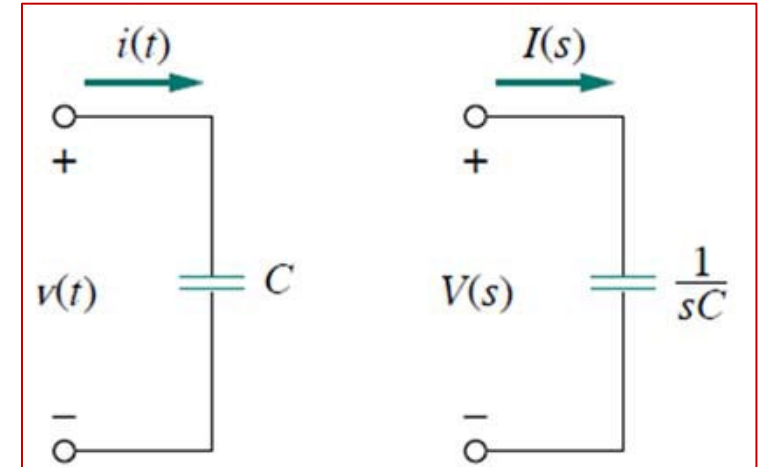
Impedance:

$$Z(s) = \frac{V(s)}{I(s)} = sL$$

Admittance:

$$Y(s) = \frac{I(s)}{V(s)} = \frac{1}{sL}$$

Capacitor



$$V(s) = I(s)/sC$$

Impedance:

$$Z(s) = \frac{V(s)}{I(s)} = \frac{1}{sC}$$

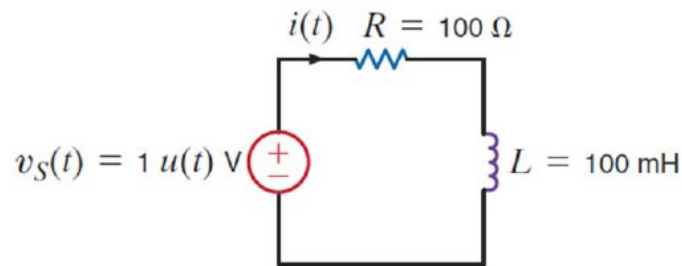
Admittance:

$$Y(s) = \frac{I(s)}{V(s)} = sC$$

Circuit Analysis in s-Domain

Example 14

- Find the current $i(t)$ for the step input applied to the RL circuit shown below:



Given: The initial value $i(0) = 0$.

Solution

The first step is to transform the circuit into the s-domain:

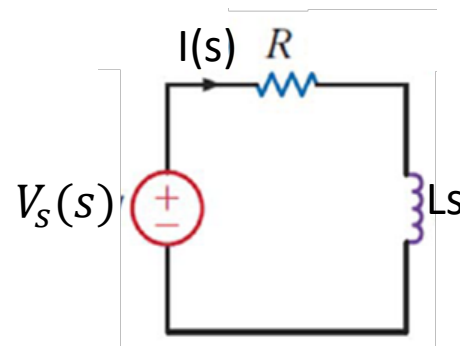
Using KVL, write the loop equation:

$$V_S(s) = R I(s) + Ls I(s)$$

We have $V_S(s) = \frac{1}{s}$

$$\text{Therefore, } I(s) = \frac{1}{s} \cdot \frac{1}{sL+R} = \frac{1/L}{s[s+\frac{R}{L}]} = \frac{A}{s} + \frac{B}{[s+\frac{R}{L}]} = \frac{1}{sR} - \frac{1}{R[s+\frac{R}{L}]}$$

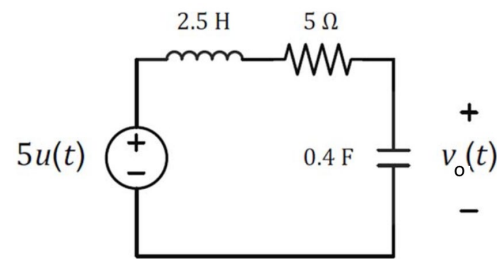
Taking the inverse Laplace transform, we obtain : $i(t) = \frac{1}{R} (1 - e^{-Rt/L}) = 10(1 - e^{-1000t}) u(t) \text{ mA}$



Circuit Analysis in s-Domain

Example 15

- Find the voltage across the capacitor $v_c(t)$ for the step input applied to the RLC circuit shown below:



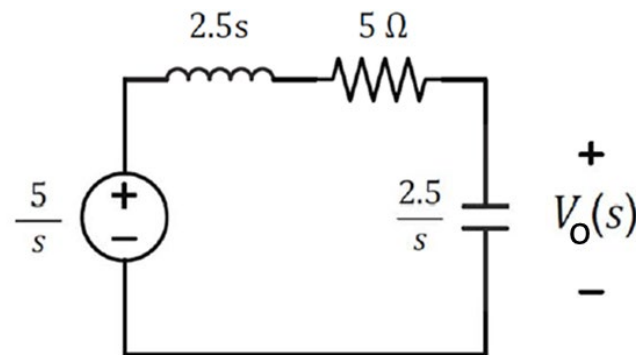
Given: The initial value $v_c(0) = 0$.

Solution

The first step is to transform the circuit into the s-domain:

Applying circuit analysis techniques, we obtain:

$$\begin{aligned} V_o(s) &= \frac{\frac{5}{s}}{2.5s + 5 + \frac{2.5}{s}} \cdot \frac{2.5}{s} \\ V_o(s) &= \frac{5}{s(s+1)^2} = 5 \left(\frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} \right) \\ V_o(s) &= \frac{5}{s} - \frac{5}{s+1} - \frac{5}{(s+1)^2} \end{aligned}$$



Taking the inverse Laplace transform, we obtain $v_o(t) = 5u(t) - 5e^{-t}u(t) - 5te^{-t}u(t) = 5(1 - e^{-t}(1 + t))u(t)$

ELEC2134: Circuits and Signals

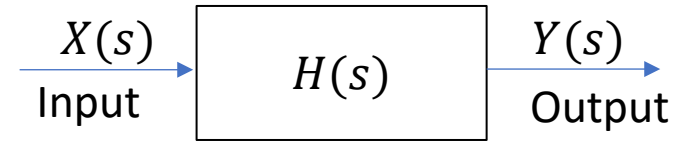
Transfer Functions

Transfer Function

- The **transfer function** is a fundamental concept in signal processing and circuit analysis. It describes how an input signal is modified as it passes through a system.
- The transfer function $H(s)$ is defined as the ratio of the output response $Y(s)$ to the input excitation $X(s)$, assuming all initial conditions are zero.

$$H(s) = \frac{Y(s)}{X(s)}$$

$$Y(s) = H(s) \cdot X(s)$$



- If both variables are voltages, the transfer function represents a **voltage gain**. If both variables are currents, it represents a **current gain**. $H(s)$ may have physical units if input and output are different physical quantities.
- If one variable is a voltage and the other is a current, the transfer function represents either an **impedance** or an **admittance**.
- There are Four Common Types:

- **1. Voltage Gain**

$$H(s) = \frac{V_o(s)}{V_i(s)}$$

- **2. Current Gain**

$$H(s) = \frac{I_o(s)}{I_i(s)}$$

- **3. Impedance**

$$H(s) = \frac{V(s)}{I(s)}$$

- **4. Admittance**

$$H(s) = \frac{I(s)}{V(s)}$$

- A single circuit may have several possible transfer functions, depending on the chosen input and output variables.
- **Note:** Gain (voltage or current) is dimensionless, whereas impedance and admittance have units.

Transfer Function – Unit Impulse Response

- To determine the transfer function $H(s)$, both the input $X(s)$ and the output $Y(s)$ must be known.
- If the input $X(s)$ and the transfer function $H(s)$ are given, the output is obtained as

$$Y(s) = H(s).X(s)$$

- Taking the inverse Laplace transform,

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

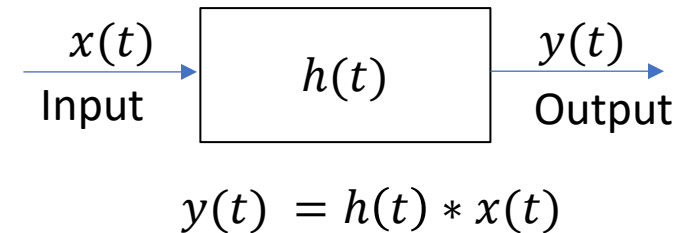
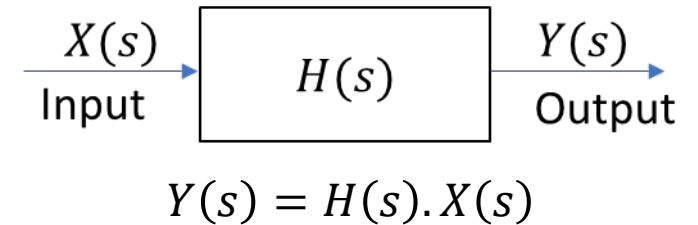
- If the input is a **unit impulse**, $x(t) = \delta(t)$, then

$$X(s) = 1$$

and therefore, $Y(s) = H(s)$ which means $y(t) = h(t)$

where

$$h(t) = \mathcal{L}^{-1}\{H(s)\}$$



- The function $h(t)$ is called the **unit impulse response**. It represents the time-domain response of the system to a unit impulse input.
- Therefore, the transfer function $H(s)$ is the **Laplace transform of the unit impulse response** of the system.

Transfer Function – Unit Impulse Response

Example 16

- If the impulse response of a network is $h(t) = e^{-t}u(t)$, determine the output $v_o(t)$ for the input $v_i(t) = 10e^{-2t}u(t)$ V.

Solution

The transformed variables are

$$\mathcal{L}[h(t)] = H(s) = \frac{1}{s+1}$$

$$\mathcal{L}[v_i(t)] = V_i(s) = \frac{10}{s+2}$$

Therefore,

$$V_o(s) = H(s) V_i(s) = \frac{10}{(s+1)(s+2)} = \frac{10}{s+1} - \frac{10}{s+2}$$

and hence,

$$v_o(t) = \mathcal{L}^{-1}(V_o(s))$$

$$v_o(t) = 10(e^{-t} - e^{-2t})u(t) \text{ V}$$

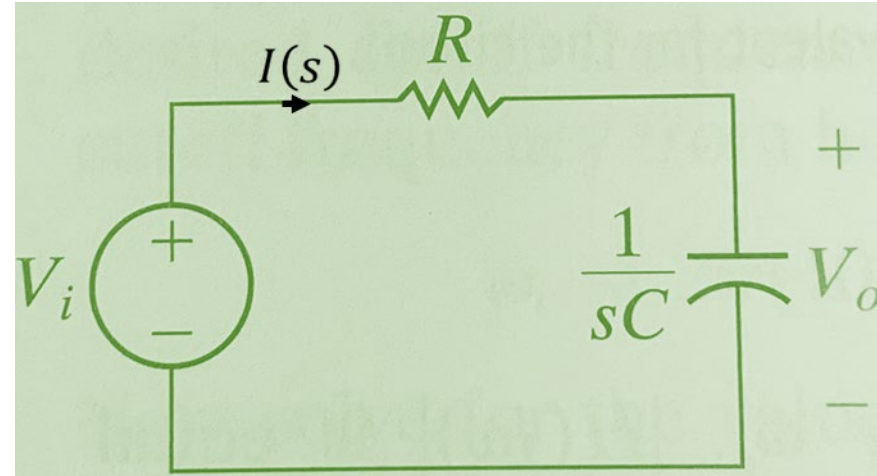
Transfer Function: Examples

Example 17

- Find the transfer function between the source voltage and the output voltage for the RC circuit given on the right:

$$I(s) = \frac{V_i(s)}{R + \frac{1}{sC}}; \quad V_o(s) = I(s) \cdot \frac{1}{sC};$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{\frac{1}{RC}}{s + \frac{1}{RC}} = \frac{a}{s+a} \quad a = \frac{1}{RC}$$

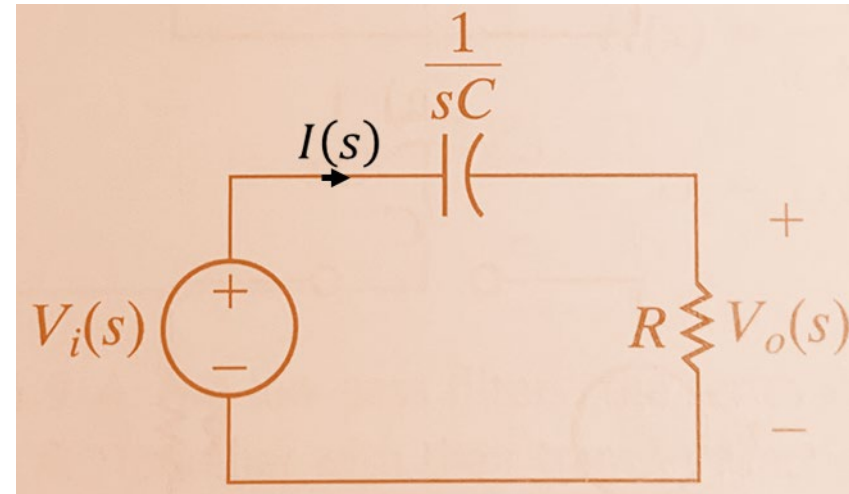


Example 18

- Find the transfer function between the source voltage and the output voltage for the RC circuit given on the right

$$I(s) = \frac{V_i(s)}{R + \frac{1}{sC}}; \quad V_o(s) = I(s) \cdot R;$$

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{R}{R + \frac{1}{sC}} = \frac{s}{s + \frac{1}{RC}} = \frac{s}{s+a}$$



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Appendix: Summary of Laplace Transform Pairs and Properties

Basic Laplace Transform Pairs and Properties

Basic Laplace Transforms

Signal	Transform
$u(t)$	$\frac{1}{s}$
$t u(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$\frac{t^{n-1}}{(n-1)!} u(t)$	$\frac{1}{s^n}$
$\delta(t - \tau), \tau > 0$	$e^{-s\tau}$
$e^{-at} u(t)$	$\frac{1}{s + a}$
$t e^{-at} u(t)$	$\frac{1}{(s + a)^2}$
$\cos(\omega t) u(t)$	$\frac{s}{s^2 + \omega^2}$
$\sin(\omega t) u(t)$	$\frac{\omega}{s^2 + \omega^2}$
$e^{-at} \cos(\omega t) u(t)$	$\frac{s + a}{(s + a)^2 + \omega^2}$
$e^{-at} \sin(\omega t) u(t)$	$\frac{\omega}{(s + a)^2 + \omega^2}$

Laplace Transform Properties

Signal	Transform
$x(t)$	$X(s)$
$ax(t) + by(t)$	$aX(s) + bY(s)$
$x(t - \tau)$	$e^{-s\tau} X(s)$
$e^{a_0 t} x(t)$	$X(s - a_0)$
$x(at), a > 0$	$\frac{1}{a} X\left(\frac{s}{a}\right)$
$x(t) * y(t)$	$X(s)Y(s)$
$-tx(t)$	$\frac{d}{ds} X(s)$
$\frac{dx(t)}{dt}$	$sX(s) - x(0)$
$\frac{d^2 x(t)}{dt^2}$	$s^2 X(s) - sx(0) - x'(0)$

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Transform Methods : Laplace Transforms Tutorial questions

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Term 3, 2026

Pass-Level Tutorial Questions (Laplace Transforms)

Question 1:

(a) Find the Laplace transform of the function $f(t) = [5 + (t + \cos 3t)e^{-2t}]u(t)$

$$\text{Ans: } \frac{5}{s} + 1/((s+2)^2 + (s+2)/(s^2+4s+13))$$

(b) Find the inverse Laplace transform of $F(s) = \frac{8}{s^3} + \frac{3}{s+2}$

$$\text{Ans: } (4t^2 + 3e^{-2t})u(t)$$

(c) Find the inverse Laplace transform of $F(s) = 5 - \frac{2}{s} + 6/(s^2 + 4)$

$$\text{Ans: } 5\delta(t) + (3 \sin 2t - 2)u(t)$$

(d) Find the Laplace transform of $f(t) = (5 + 6te^{-3t} - 4e^{-t} \cos 2t)u(t)$.

$$\text{Ans: } F(s) = \frac{5}{s} + 6/((s+3)^2 - 4(s+1)/\{(s+1)^2 + 2^2\})$$

Then use the initial – value and final – value theorems to check your results .

(e) Find the inverse Laplace transform of $F(s) = \frac{2s^2+s-3}{s(s^2+4s+3)}$

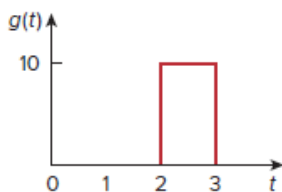
$$\text{Ans: } f(t) = (-1 + 1e^{-t} + 2e^{-3t})u(t)$$

(f) Find : (i) $\mathcal{L} \left\{ \frac{d}{dt} (\sin \omega t) \right\}$; (ii) $\mathcal{L} \left\{ \frac{d}{dt} (\cos \omega t) \right\}$; (iii) $\mathcal{L} \left\{ \frac{d^3}{dt^3} (t^2) \right\}$

$$\text{Ans: (i) } \frac{s\omega}{s^2+\omega^2}; \text{ (ii) } \frac{-\omega^2}{s^2+\omega^2}; \text{ (iii) } 0$$

Check the results of parts(a), (b) and (c) by first differentiating and then transforming.

(g) Determine the Laplace transform of the following function:



$$\text{Ans: } \frac{10}{s} (e^{-2s} - e^{-3s})$$

(h) Determine the Laplace transform of each of the following functions:

$$\text{(i) } f(t) = (\cos(2t) + e^{-4t})u(t)$$

$$\text{Ans: } \frac{2s^2+4s+4}{(s^2+4)(s+4)}$$

$$\text{(ii) } f(t) = t^2 \sin 2t u(t)$$

$$\frac{12s^2 - 16}{(s^2 + 4)^3} \quad \text{s:}$$

Pass-Level Tutorial Questions (Laplace Transforms)

Question 2: Given the following differential equation, solve for $y(t)$ if all initial conditions are zero.

$$\frac{d^2 y(t)}{dt^2} + 12 \frac{dy(t)}{dt} + 32 y(t) = 32 u(t)$$

Ans: $y(t) = (1 - 2e^{-4t} + e^{-8t})$

Question 4: Consider a system with the transfer function $H(s) = \frac{1000(s+4)}{(s+5)(s^2+8s+32)}$. Suppose that the system input is $x(t) = e^{-t}\{1 - e^{-3t}\}u(t)$, Find the response $Y(s)$ of the system.

Ans: $Y(s) = \frac{3000}{(s+1)(s+5)(s^2+8s+32)}$

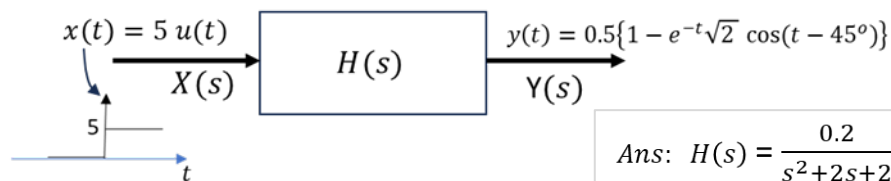
Question 6: The relationship between the input $x(t)$ and output $y(t)$ of a system is described by the following differential equation:

$$\frac{d^2 y(t)}{dt^2} - 2 \frac{dy(t)}{dt} + 10 y(t) = x(t) + 2 \frac{dx(t)}{dt}$$

Use Laplace transforms to determine the transfer function $H(s) = Y(s)/X(s)$ and show the impulse response of the system $h(t)$ is given by

$$h(t) = e^t \{2 \cos(3t) + \sin(3t)\} u(t)$$

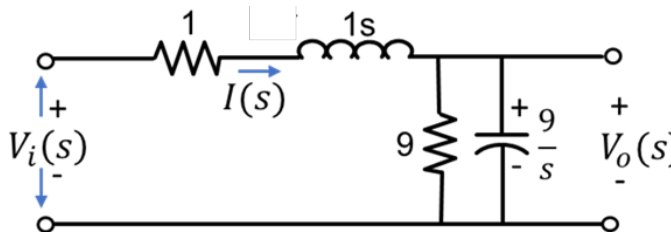
Question 3: Determine the transfer function of the system shown below:



Ans: $H(s) = \frac{0.2}{s^2 + 2s + 2}$

Question 5: An RLC circuit is given below.

- Find the transfer function of the circuit.
- Find the circuit response to a unit-step input.
- Find the initial and final values of $v_o(t)$
- Find the impulse response of the system and sketch $h(t)$ with reasonable detail.



Ans:

(a) $H(s) = \frac{9}{(s^2 + 2s + 10)}$

(b) $V_o(s) = \frac{9}{(s^2 + 2s + 10)} \cdot \frac{1}{s}$

(c) Initial value = 0;
Final Value = 0.9

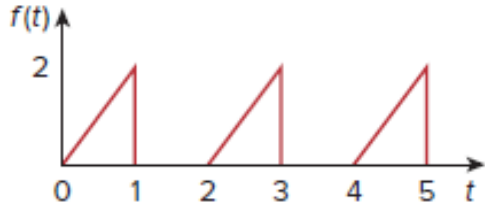
(d) $h(t) = e^{-t} \sin(3t) u(t)$

Question 7: If the impulse response of a circuit is $h(t) = 10e^{-2t}u(t)$, what is its response to a unit step?

Ans: $y(t) = 5(1 - e^{-2t})u(t)$

Higher-Level Tutorial Questions (Laplace Transforms)

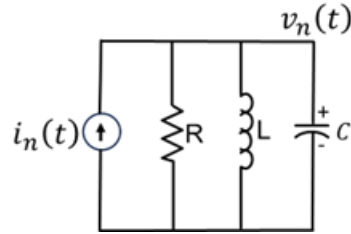
Question 1: Determine the Laplace transform of the following function:



Ans:
$$F(s) = \frac{2}{s^2(1 - e^{-2s})}(1 - e^{-s} - se^{-s})$$

Question 2: For the circuit shown below, (i) Show that the relationship between the input current $i_n(t)$ and the output voltage $v_n(t)$ is given by the following differential equation. You may assume zero initial conditions.

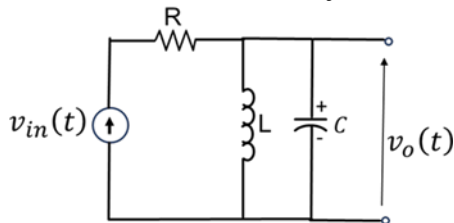
$$\frac{d^2 v_n(t)}{dt^2} + \frac{1}{RC} \frac{dv_n(t)}{dt} + \frac{1}{LC} v_n(t) = \frac{1}{C} \frac{di_n(t)}{dt}$$



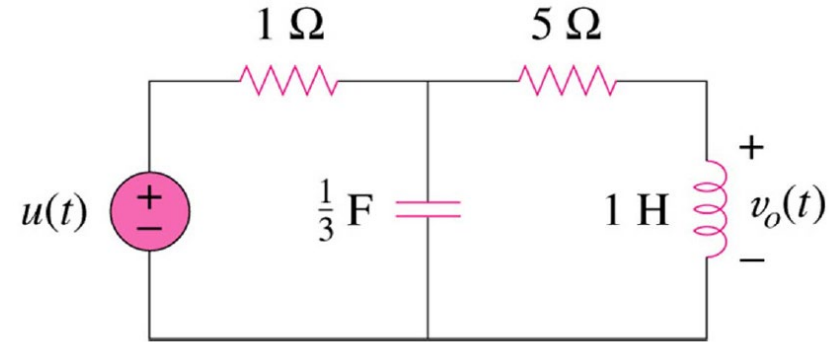
(ii) Use Laplace transforms to determine $\frac{V_n(s)}{I_n(s)}$ and show $\frac{V_n(s)}{I_n(s)} = \frac{\frac{1}{C}s}{(s^2 + \frac{1}{RC}s + \frac{1}{LC})}$

Question 3: The Figure shown below is a RLC based circuit: (i) Show that the

transfer function $H(s) = V_o(s)/V_{in}(s) = \frac{\frac{s}{RC}}{s^2 + \frac{\omega_o}{Q}s + \omega_o^2}$ where $\omega_o = 1/\sqrt{LC}$ and $Q = \omega_o RC = R/L\omega_o$;



Question 4: Find $v_o(t)$ for the circuit shown below, assuming zero initial conditions, by transforming the circuit from the time domain to the s-domain.

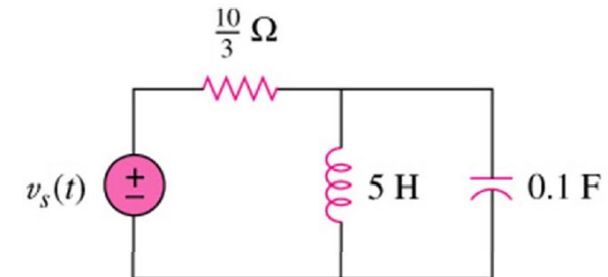


Ans:
$$v_o(t) = \frac{3}{\sqrt{2}} e^{-4t} \sin(\sqrt{2}t) \text{ V}, \quad t \geq 0$$

Question 5: Find the voltage across the capacitor, given that

$$v_s(t) = 10 u(t) \text{ V},$$

and assuming that at $t = 0$, a current of -1 A flows through the inductor and $+5 \text{ V}$ is across the capacitor.



Ans:
$$(35e^{-t} - 30e^{-2t}) u(t) \text{ V}$$